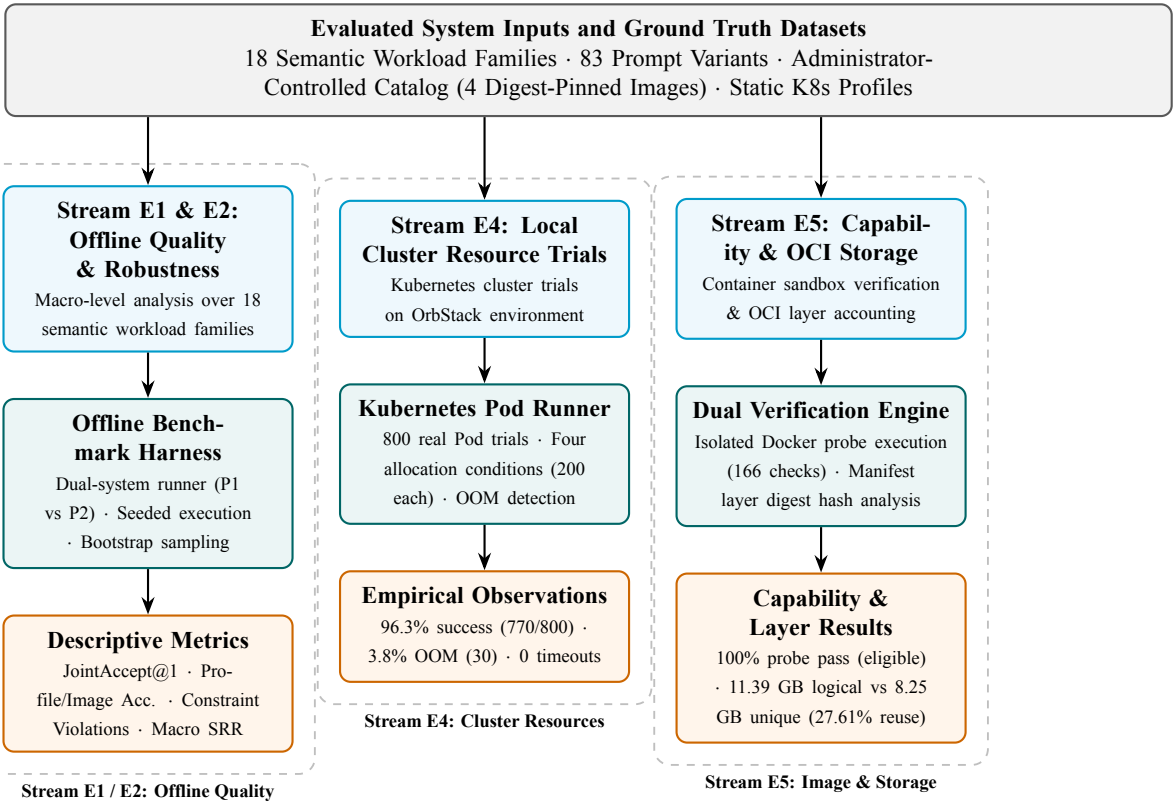

Evaluated System Inputs and Ground Truth Datasets
18 Semantic Workload Families · 83 Prompt Variants · Administrator-Controlled Catalog (4 Digest-Pinned Images) · Static K8s Profiles
Stream E1 & E2: Offline Quality & Robustness
Macro-level analysis over 18 semantic workload families
Offline Benchmark Harness
Dual-system runner (P1 vs P2) · Seeded execution · Bootstrap sampling
Descriptive Metrics
JointAccept@1 · Profile/Image Acc. · Constraint Violations · Macro SRR
Stream E1 / E2: Offline Quality
Stream E4: Local Cluster Resource Trials
Kubernetes cluster trials on OrbStack environment
Kubernetes Pod Runner
800 real Pod trials · Four allocation conditions (200 each) · OOM detection
Empirical Observations
96.3% success (770/800) · 3.8% OOM (30) · 0 timeouts
Stream E4: Cluster Resources
Stream E5: Capability & OCI Storage
Container sandbox verification & OCI layer accounting
Dual Verification Engine
Isolated Docker probe execution (166 checks) · Manifest layer digest hash analysis
Capability & Layer Results
100% probe pass (eligible) · 11.39 GB logical vs 8.25 GB unique (27.61% reuse)
Stream E5: Image & Storage